

Viral kinetics in the COVID-19 pandemic

COVID-19 testing: One size does not fit all

To control the pandemic, testing should be considered a public health tool

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COVID-19 testing strategies

Testing for SARS-CoV-2 can be for personal or population health. Collection can be from symptomatic or asymptomatic individuals, as well as from wastewater and swabs of surfaces. The tests may be performed in central laboratories, at the POC, or using rapid tests. Attributes of tests differ according to application.

Personal health

Diagnostics

Symptomatic
At-risk

Testing

qPCR, LAMP, Ag

Key attributes

Sensitivity 
Specificity 
Frequency 

Population health

Surveillance

Asymptomatic
Wastewater
Surfaces

Testing

qPCR, Ag, LAMP
Ab, pooling

Key attributes





Screening

Everyone
Entrance Public health

Ag plus confirmation with Ag,
LAMP, or qPCR; LAMP











CU Boulder Today

<https://www.colorado.edu/today/2020/11/20/cu-making-saliva-based-pcr-covid-19-tests-available-immediate-family-household-members>



census 2021 [Data and analysis from Census 2021](#)

Coronavirus (COVID-19) Infection Survey, UK: 24 March 2023

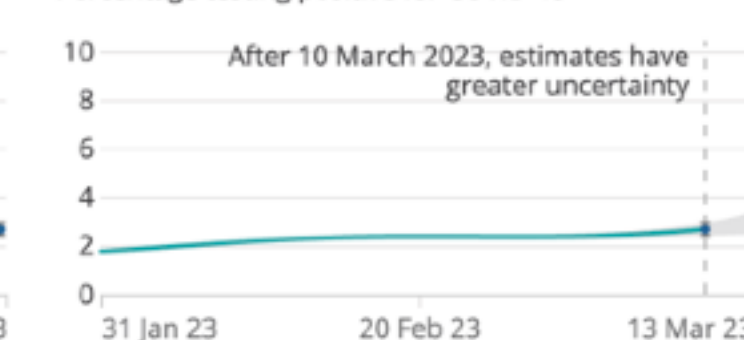
Percentage of people testing positive for coronavirus (COVID-19) in private residential households in England, Wales, Northern Ireland and Scotland, including regional and age breakdowns. This survey is being delivered in partnership with University of Oxford, University of Manchester, UK Health Security Agency (UKHSA) and Wellcome Trust, working with the University of Oxford and partner laboratories to collect and test samples.

England

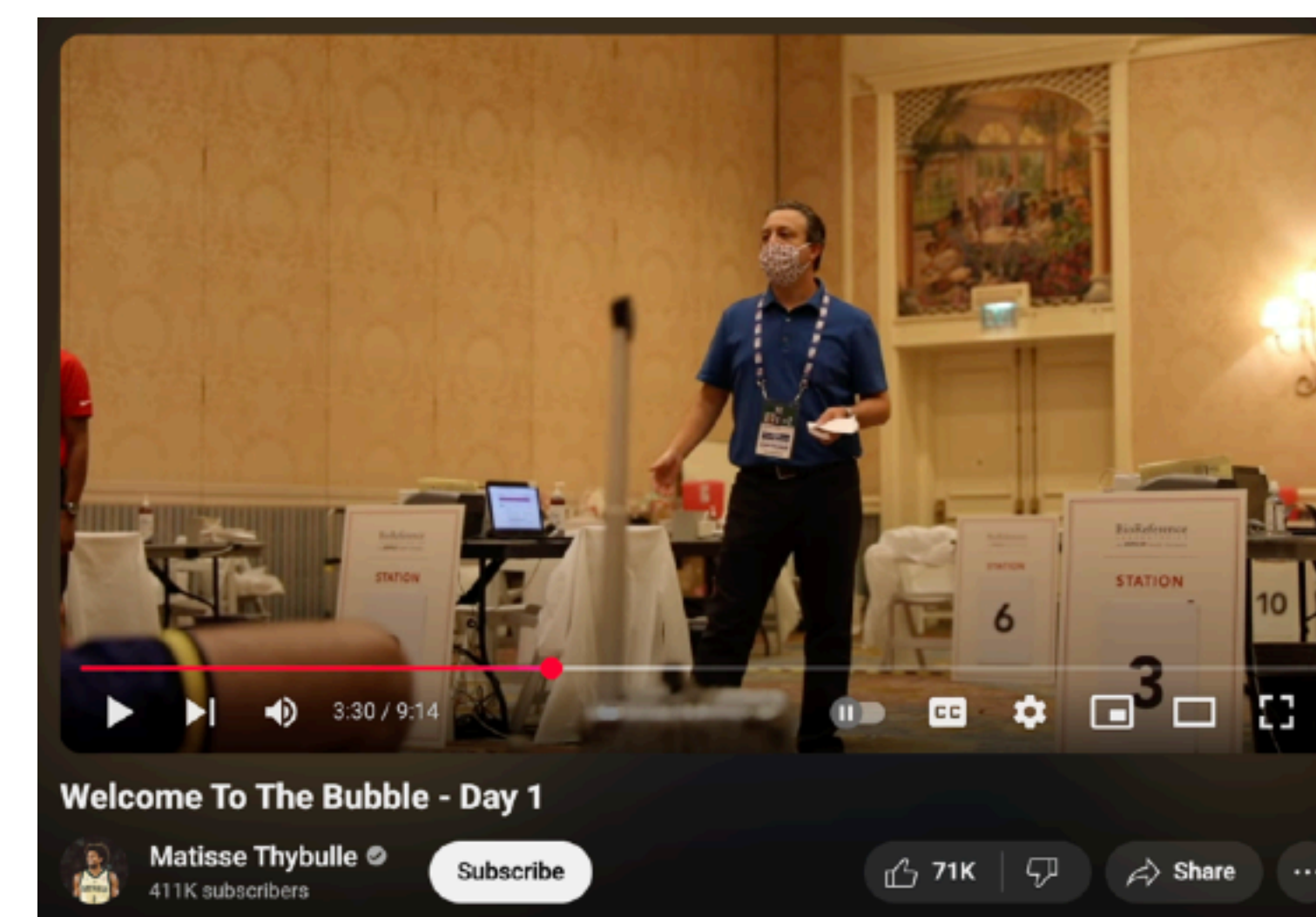
Percentage testing positive for COVID-19



Percentage testing positive for COVID-19



<https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/conditionsanddiseases/bulletins/coronaviruscovid19infectionsurveys/pilot/latest>



https://www.youtube.com/watch?v=_DzaVY5tPZg

Viral kinetics data during the COVID-19 pandemic

Targeted studies

Large prospective studies

Occupational health programs